

Report

This smart home application is a simple Python-based prototype that demonstrates how natural language can be used to control and monitor household devices through Generative AI. The project combines simulated smart devices with a Hugging Face Transformers model so that users can interact with the system in a more intuitive way. Instead of clicking buttons or entering rigid commands, a user can type requests such as “Turn on the light,” “Set the fan speed to medium,” or “What is the current temperature?” The application interprets the meaning of the request, converts it into a structured internal command, executes the appropriate action, and then returns a clear response.

At the core of the system are three simulated devices: a light, a fan, and a thermostat. Each device is represented as a Python class and maintains its own internal state. The light supports basic ON/OFF control and can report whether it is currently on or off. The fan supports ON/OFF control as well as three speed settings: low, medium, and high. Its behavior has an important safeguard: the fan’s speed can only be changed when the fan is already turned on. If the fan is off, it does not maintain an active running speed, and the system will return an error message if a user attempts to change speed while it is off. The thermostat allows the temperature to be adjusted within a defined range between 18°C and 30°C and can also report the current setting.

The AI component is implemented using Hugging Face Transformers, specifically a zero-shot classification pipeline based on the `facebook/bart-large-mnli` model. This model is used to infer the intent behind a natural language command. The parser classifies user input into categories such as turning a device on, turning a device off, setting fan speed, setting temperature, or requesting device status. After intent classification, the system applies lightweight extraction rules to identify which device is being addressed and whether a value such as “high” or “24” is part of the instruction. The result is a structured command object containing fields like the raw input, detected intent, target device, action, value, and model confidence score.

Once a structured command has been created, the smart home controller maps it to the correct device action. This controller is the part of the application that translates AI interpretation into actual system behavior. For example, if the parsed command indicates that the light should be turned on, the controller calls the light’s `turn_on()` method and then returns a response such as “The light is now ON.” If the user sets the fan speed to medium while the fan is on, the system responds with “The fan speed is set to medium.” If the user adjusts the thermostat to 24 degrees, the confirmation becomes “The thermostat is set to 24°C.” Status-related queries are also supported, so a user can ask about the current temperature or the state of a device and receive a concise answer.

The project also includes a demo mode that shows the entire flow from end to end. When run, it prints the original user input, the structured command generated by the parser, the feedback returned to the user, and the updated device status. In addition, an interactive command-line interface is available for live demonstrations. This CLI allows users to type commands naturally in a loop, view current device states, request help with sample prompts, and optionally display the structured command output for more technical presentations.

Overall, the application serves as a clear example of how Generative AI can be integrated into a smart home environment. It shows how natural language understanding, structured command parsing, and device simulation can work together in a practical and extensible

way. While the current version is intentionally simple and runs entirely in memory, it provides a strong foundation for future improvements such as supporting multiple rooms, adding more device types, building a web interface, or introducing persistent state management.